

Program 6

Deploy an Automated Build Pipeline using Docker Hub

The program aims to deploy an automated build pipeline using Docker Hub for continuous image building and distribution. This ensures that every code push to the repository triggers Docker Hub to build and store updated container images.

Project Structure:

program_6/

—>Dockerfile

—>[app.py](#)

—> requirements.txt

Procedure:

Step 1: Create a simple **Dockerfile** without any extensions, Create a simple [app.py](#) file, Create a simple requirements.txt file, Build an image and Run the container

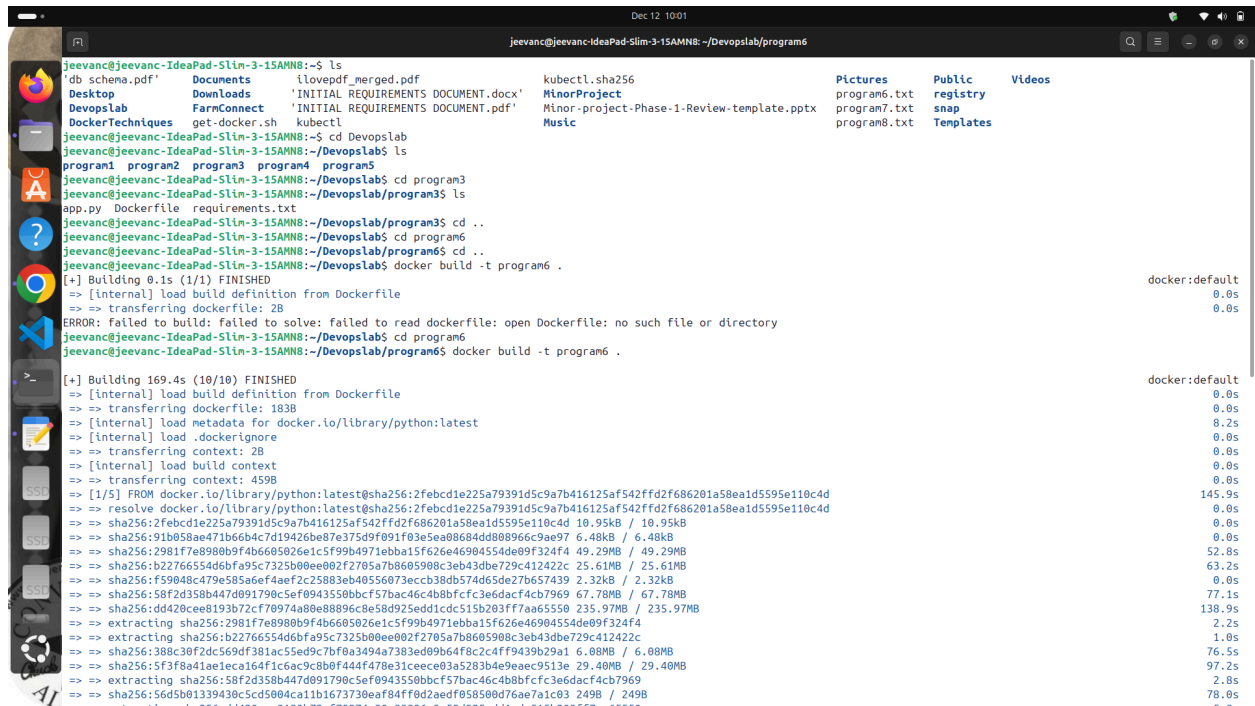
```
nano Dockerfile
```

```
nano app.py
```

```
nano requirements.txt
```

```
docker build -t program_6 .
```

```
docker run -p 5000:5000 program_6
```



Step 2: Tag the created image for Docker Hub

`docker tag program_6 <dockerhub-username>/program_6:latest`

Step 3: Tag the Image for GitHub Container Registry (GHCR) and Push the Image to GHCR

`docker tag program_6 ghcr.io/github-username/program_6:latest`

`echo $GITHUB_PAT | docker login ghcr.io -u <github-username> --password-stdin`

Step 4: Push source code to GitHub

`git add .`

`git commit -m "Initial Commit"`

`git push -u origin main`

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program6
=> extracting sha256:38bc30f2dc569df381ac55ed9c7bfa3494a7383ed09b64f8c2c4ff9439b29a1 0.2s
=> extracting sha256:5f3f8a41ae1ca164f1c6ac9c8b0f444f478e31ceec03a5283b4e9eac9513e 0.6s
=> extracting sha256:56d5b01339430c5cd5004ca11b1673730eaf84ff0d2aedf058500d76ae7a1c03 0.0s
=> [2/5] WORKDIR /app 1.0s
=> [3/5] COPY requirements.txt . 0.0s
=> [4/5] RUN pip install -r requirements.txt 13.7s
=> [5/5] COPY . . 0.1s
=> exporting to image 0.2s
=> exporting layers 0.2s
=> writing image sha256:462428d21b9ad9819d4d9dcde9c305f3135f7c82c271cca8df3a87cf4e47b76 0.2s
=> naming to docker.io/library/program6 0.0s
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program6$ docker tag program6 jeevanc370/program6:latest
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program6$ docker login -u jeevanc370
Info -> A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/home/jeevanc/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program6$ docker push jeevanc370/program6:latest
The push refers to repository [docker.io/jeevanc370/program6]
4a1feeaa2156: Pushed
079f5fd8e072: Pushed
66e11591956a: Pushed
fff354dbd22f: Pushed
99026168d02f: Mounted from library/python
5c1d27353faa: Mounted from library/python
f89b412f0854: Mounted from library/python
662b8975365c: Mounted from library/python
8e47c7b8acf3: Mounted from library/python
0b288bc29d30: Mounted from library/python
e9b018e49a06: Mounted from library/python
latest: digest: sha256:e3de73333aff349e0a0760477b0a840ae86dd8e22c2069569d05772dc068add7 size: 2627
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program6$ docker tag program6 jeevanc370/program6:latest
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program6$ export GITHUB_PAT=ghp_E9FRZlWVNmPkRQ181YnB3Ujv1XGLHdCGQj
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program6$ echo $GITHUB_PAT | docker login -u jeevanc370
Error: Cannot perform an interactive login from a non TTY device
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program6$ docker tag program6 jeevanc370/program6:latest
```

